Kick-off Diagnostic · L6-6 — Exponential and Logarithmic Equations

Grade 11

8 questions · 30 seconds each · about 4 minutes · Diagnose phase (6 min of 60)

Pre-session anchor (slides 2 and 3 of the pre-session must say this, so items 2 and 6 test it).
The one idea: Write both sides with one base and equate the exponents; if the bases cannot match, take a logarithm of both sides. Check every log solution in the original equation.
The mistake most students make: Dividing 32 by 4 to solve $4^x = 32$ and answering $x = 8$ instead of $x = \frac{5}{2}$.
Readiness items 3, 4, 5, 7, 8 test: powers of 2, 3 and 4 ($4 = 2^2$), $\log_b x$ as an exponent (Lesson 6-3), reading graphs of b^x and $\log_b x$.
Wayground set-up: timer 30 s on every question; shuffle options on, questions off; hide the leaderboard; show answers after the quiz, not after each item. Item 1 is a poll (no points). Items 2 to 8 are 1 point each. Tags below: **maroon** = tells you about the pre-session, **navy** = readiness for today. **Green** = correct.

1 POLL · 30 s — SELF-REPORT

Be honest: how much of the pre-session task did you do?

A. All six stations, including the checks
B. I watched and read but skipped the checks
C. I opened it but did not finish
D. I did not open it

Signal. Ungraded. Tally it first; it is the self-report half of the pre-session test.

2 MULTIPLE CHOICE · 30 s — PRE-SESSION CHECK

In the pre-session, when can you solve $b^x = c$ by equating exponents?

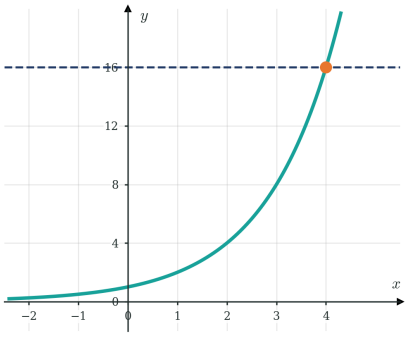
✓ **A. When both sides can be written with the same base**
B. Only when c is an even number
C. When x is a whole number
D. Always, by dividing c by b

Signal. Wrong → did not watch station 4. Picking D = the pre-session's warned mistake, not a method.

3 FILL IN THE BLANK · 30 s — READINESS

The graphs of $y = 2^x$ and $y = 16$ are shown. The solution of $2^x = 16$ is $x =$ _____

Accept: 4



Signal. 8 → divided 16 by 2; 16 → read the y-value. Solving by inspection is not yet secure.

4 CHECKBOX · 30 s — READINESS

Select the **two** true statements.

✓ **A.** $4 = 2^2$
B. $27 = 3^2$
✓ **C.** $8 = 2^3$
D. $9 = 3^3$

Signal. Choosing $27 = 3^2$ or $9 = 3^3$ → powers of 2 and 3 are not secure; matching bases will stall.

5 MATCH · 30 s — READINESS

Match each logarithm with its value.

$\log_2 16$ → 4
$\log_3 27$ → 3 (shown shuffled in Wayground)
$\log_{10} 100$ → 2

Signal. Any miss → the log-to-exponent link (Lesson 6-3) needs a quick re-teach.

6 MULTIPLE CHOICE · 30 s **PRE-SESSION CHECK**

The pre-session warned about solving $4^x = 32$. Which wrong answer was it?

✓ **A.** $x = 8$ **(from $32 \div 4$)**
B. $x = \frac{5}{2}$
C. $x = 2$
D. $x = 4$

Signal. The pre-session's "mistake most students make". Wrong → skimmed station 4.

7 REORDER · 30 s **READINESS**

Put the steps for solving $4^x = 32$ in order.

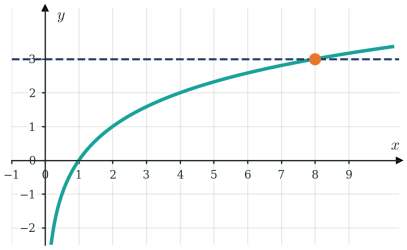
Correct order: $4^x = 32 \rightarrow (2^2)^x = 2^5 \rightarrow 2x = 5 \rightarrow x = \frac{5}{2}$

Signal. Wrong → rewriting in one base is the gap; start the lesson with $4 = 2^2$.

8 DROPDOWN · 30 s **READINESS**

The graph shows $y = \log_2 x$ and $y = 3$. The solution of $\log_2 x = 3$ is $x =$...

A. 3
B. 6
✓ **C.** 8
D. 9



Signal. $6 = 2 \times 3$; $9 = 3^2$; 3 = read the y-value. Converting log form to exponent form is the gap.

Read the results (90 seconds)

Readiness R — **Items 3, 4, 5, 7, 8 (out of 5). 0–2** → re-teach matching bases and log values with me. **3–4** → straight to solving equations; fix the missed item in passing. **5** → start the Investigate prompt now.

Pre-session P — **Items 2 and 6 (out of 2), read with the item 1 poll.**

What you see	Reading	What you do
Poll "all" or "read", $P = 2$	Prepared	Nothing extra. Use their R score to route.
Poll "all" or "read", $P = 0$	Claimed, did not retain	Two-minute recap of the one idea at the start. Ask them to restate it in their own words.
Poll "opened" or "did not open"	Task not done	No penalty. Pair with a prepared partner for Targeted Instruction; note it for your task tracker.
$R \leq 2$ and $P = 2$	Content gap	Re-teach group. The task was done, so the idea needs a different explanation.
$R \geq 4$ and $P \leq 1$	Task skipped, skills fine	Move on. Chase the task, not the content.

Class-level rule. If more than 30% of the class chose "opened" or "did not open", or the class average on P is below 1, spend three minutes recapping the one idea above before Targeted Instruction. If more than 30% score $R \leq 2$, the whole class takes the re-teach route first.